

Evaluating Missing Data Handling Techniques Under Different Missingness Mechanisms

Rophence Ojiambo

Kiran Nagdeo

Cindy Patippe

Introduction

Missing data, whether due to corruption or logistical issues, can bias estimates, inflate standard errors, and invalidate inferences if not properly handled^[1]. Rubin's (1976) framework [2] defines three mechanisms: Missing Completely at Random (MCAR), where missingness is unrelated to any data; Missing at Random (MAR), where missingness depends only on observed data; and Missing Not at Random (MNAR), where missingness depends on unobserved values, posing the greatest challenge and requiring strong, often unverifiable assumptions [3]. Complete-case analysis is commonly used in applied research for its simplicity, but under MAR or MNAR conditions, it can introduce bias and reduce efficiency [4]. Single imputation methods (e.g., mean substitution) also perform poorly, distorting distributions and underestimating uncertainty. In contrast, Multiple Imputation by Chained Equations (MICE) and related model-based approaches simulate missing values multiple times and pool results to reflect imputation uncertainty [5]. While increasingly popular, the performance of these methods varies depending on the nature and extent of missingness mechanism [6]. Simulation studies allow systematic evaluation of these methods by generating complete data, imposing missingness mechanisms, and comparing outcomes such as bias, confidence interval coverage, prediction accuracy, and model fit (e.g., AIC, BIC) [7 ; 8].

Relevance to Obesity Research

Obesity presents a growing public health challenge worldwide, contributing to increased risk of cardiovascular disease, type 2 diabetes, hypertension, musculoskeletal disorders, and certain cancers. According to the World Health Organization (2024)[9], global obesity prevalence has tripled since 1975, and over 650 million adults are currently classified as obese. In Latin America, where countries like Mexico, Peru, and Colombia are experiencing rapid nutrition and lifestyle transitions, obesity rates have surged across all age groups[10; 11; 12]. The determinants of obesity are multifactorial, including sociodemographic variables, poor dietary patterns, inadequate physical activity, sedentary behaviors, and harmful habits such as smoking and alcohol consumption[13;14]. These risk factors are often collected via self-report or web-based tools, making them especially vulnerable to missingness due to social desirability bias, recall error, or issues with technology access. Understanding how missing data may bias the estimation of such relationships is critical for valid epidemiologic inference.

Study Objective

The objective of this project is to conduct a simulation study using the [UCI Obesity](#) dataset to assess how various imputation techniques perform under MCAR, MAR, and MNAR scenarios.

Methods

Data Source

We used the `obesity.levels` dataset from the `fairml` R package, which contains 2,111 records from individuals in Mexico, Peru, and Colombia. The dataset includes 17 demographic, behavioral, and physical condition attributes used to estimate obesity levels, labeled by the `NObeyesdad` variable. Approximately 77% of the data were synthetically generated using Weka and SMOTE, while 23% were collected via a web platform. Further details are elsewhere^[2,3]. The binary outcome variable, denoted as `obese_binary`, was derived from the original multiclass label `NObeyesdad` by coding individuals with Obesity Type I, II, or III as “Obese” and others as “Not Obese”. Continuous predictors included: Age, FCVC (Frequency of vegetable consumption), NCP (Number of main meals), CH2O (daily water intake), FAF (physical activity frequency), and TUE (time using technology). Categorical variables included Gender, family history with overweight, high-calorie food consumption (FAVC), food between meals (CAEC), smoking (SMOKE), calorie monitoring (SCC), alcohol consumption (CALC), and transportation type (MTRANS).

Missing Data Generation Mechanisms

We simulated missingness under three mechanisms: To simulate MCAR, we randomly introduced missing values in three variables; `Gender`, `FAF`, and `CH2O`, with predefined rates of 15%, 25%, and 20%, respectively. The probability of missingness in these variables is independent of any observed or unobserved data. We simulated MAR by conditioning missingness on observed covariates. Specifically: `FAF` was more likely to be missing for older males; `SCC` was more likely to be missing among participants reporting high-calorie food consumption; `NCP` was more likely to be missing for younger females. Probabilities of missingness were generated using logistic functions of these predictors. For MNAR, missingness depended on unobserved (true) values of the variables themselves: `FAF` missingness was more likely in individuals with low physical activity; `CALC` was more likely missing in frequent drinkers; `SCC` missingness was higher among participants who did not monitor calories. These dependencies reflect real-world challenges where values themselves influence their probability of being missing.

Imputation Models

We applied three common strategies to handle missing data; (1) Complete Case Analysis where all records with missing values in any of the selected predictors were excluded; (2) Mean/Mode Imputation where for continuous variables, missing values were imputed using the mean of observed values and for categorical variables, the most frequent category (mode) was used; (3) MICE using the `mice` package was used to perform 5 multiple imputations using chained equations.

Model Fitting and Evaluation

We trained logistic regression models to predict `obese_binary` across all datasets (Original, MCAR, MAR, MNAR) and imputation methods. Each dataset was split into training (70%) and test (30%) sets using stratified sampling to preserve the outcome distribution. Model performance was evaluated using the following metrics: **Accuracy**: Proportion of correct classifications; **Sensitivity (Recall)**: Proportion of obese individuals correctly identified; **Specificity**: Proportion of non-obese individuals correctly identified; **Area Under the ROC Curve (AUC)**: Discriminative ability of the model; **F1 Score**: Harmonic mean of precision and recall. Model metrics were computed separately for the training and test sets. For MICE, pooled predictions across imputations were used for test metrics, and average metrics were reported for training.

Results

Data Availabiliy

All codes and datasets for this project can be found at our [GitHub Repository](#).

References

1. Rubin, D. B. (1987). *Statistical analysis with missing data*. Wiley.
2. Palechor, F. M., & De la Hoz Manotas, A. (2019). Dataset for estimation of obesity levels based on eating habits and physical condition in individuals from colombia, peru and mexico. *Data in Brief*, 25, 104344.
3. De-La-Hoz-Correa, E., Mendoza Palechor, F., De-La-Hoz-Manotas, A., Morales Ortega, R., & Sánchez Hernández, A. B. (2019). *Obesity level estimation software based on decision trees*.